

Etiologies

↓ Effective Arterial Blood Volume (EABV)

1. Hypovolemia
2. Sepsis
3. Heart failure
4. Cirrhosis

Altered vasculature

1. Renal vein thrombosis
2. Renal vein HTN (CHF)
2. Intra-Abdominal Hypertension (IAH)

Dx

Suggestive history, bland urinalysis, no hydronephrosis and FeNa <1% (if oliguric)

PRE-renal

Click here for additional details

Etiologies

Post-renal AKI manifests through an **obstructive nephropathy**. If the outflow of **BOTH kidneys** (or the single functioning kidney) anywhere from the level of the urethra up to the renal calyx then: ↑ retrograde hydrostatic pressure, and resultant ↓ GFR.

Urethra -> Bladder -> Ureter

- Stricture	- Clot	- Calculi
- BPH	- Malignancy	- Int/ext malignancy or LAN
	- Neurogenic	- Retroperitoneal fibrosis

D

× Suggestive history and hydronephrosis on ultrasound of kidney

POST-renal

Click here for additional details

Bladder

Prostate

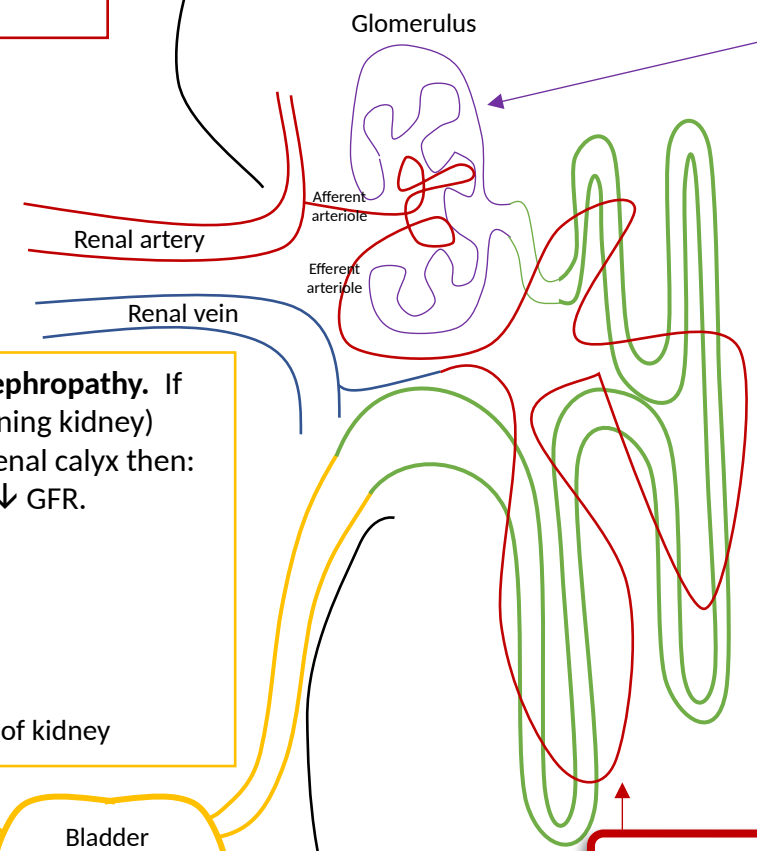
Acute Kidney Injury

(Abrupt rise in serum Cr)

Key point: eGFR is not meaningful in setting of AKI. eGFR is based on assumption of stable GFR

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INTRA-renal

Broad differential broken down anatomically by **glomerular**, **tubulointerstitial** and **vascular**.

Dx

1. "Active" urine sediment (supports intra-renal)
2. Elevated FeNa while oliguric (rules out pre-renal)
3. Normal kidney US (rules out post-renal).

Glomerular

Acute Glomerulonephritis (GN): both **non-crescentic** (e.g. class IV lupus nephritis or infection-related) and **crescentic GN** or "**RPGN**" (grouped into immune-complex, pauci-immune and anti-GBM).

Dx: dysmorphic RBCs / RBC casts, proteinuria, leukocyturia, +/- HTN

Tubulointerstitial

Tubular

- **Acute tubular necrosis (ATN)** *most common*: 1) ischemic (prolonged hypoperfusion) or 2) toxic (including iodinated contrast)
- **Tubular obstruction** 1) cast nephropathy (pigments - myoglobin/hemoglobin or proteins - light-chains/paraproteins) or 2) crystals (uric acid, acyclovir, ethylene glycol, MTX).

Acute interstitial nephritis

- **Drugs** - antimicrobials, NSAIDs, PPIs, immunotherapy
- **Infection** - pyelonephritis, *Legionella*, *Leptospira*, CMV
- **Systemic diseases** - Sjogren's, sarcoidosis

Vascular

Vasogenic (change in tone of afferent or efferent arterioles)

- Constriction of afferent - e.g. NSAIDs, calcineurin inhibitors (CNIs)
- Dilation of efferent arteriole - e.g. ACE inhibitors, ARBs

Microangiopathic Hemolytic Anemia (MAHA)

- TTP/HUS, scleroderma renal crisis, malignant HTN, HELLP).

Cholesterol Emboli

Click

Case 1

35yo woman presenting to ED with one week of vomiting and diarrhea. Reports decreased UOP and light headedness

Cr **1.5** (baseline 0.7) | BUN **45** | FeNa <0.2%
Urinalysis: hyaline casts | Kidney US: Normal

Pre-renal azotemia from hypovolemia

Control nausea, aggressive rehydration

Case 2

70yo man presenting to clinic with difficulty initiating urination, weak stream and incomplete emptying for the past couple of years that has significantly worsening over the past week since he started taking allergy medications.

Cr **2.0** (baseline 0.8) | BUN **40** | FeNa 1.5%
Urinalysis: nml | Kidney US: Bilateral mod hydronephrosis

Post-renal nephropathy from BPH + medication induced urinary retention.

Discontinue anti-cholinergic, start tamsulosin, consider urinary catheter

Case 3

23yo transman with SLE not currently on therapy presents to the ED with arthralgias, dark urine, decreased urine output.

Cr **3.6** (baseline 0.9) | BUN 30 | FeNa 2.8%
Urinalysis: dysmorphic RBCs and RBC casts
Kidney US: Normal

Likely acute glomerular nephritis from lupus flare

Induction therapy with steroids + cyclophosphamide or mycophenolate

Case 4

40yo woman returns to clinic with decreased UOP and fatigue a 5 days into a 14 day course of amoxicillin/clavulanate for acute sinusitis. Noted to have a faint truncal maculopapular rash.

Cr **2.5** (baseline 0.9) | BUN 30 | FeNa 2.2%
Urinalysis: proteinuria, WBCs & WBC casts. No eosinophils
Kidney US: Normal

Likely acute interstitial nephritis (AIN) from beta-lactam

Discontinue abx, repeat labs in 3 days and consider bx +/- steroids if not improving

Case 5

42yo man with DMII is admitted for pneumonia with severe sepsis and hyperglycemia. He received aggressive IVF resuscitation but his Cr continues to rise 24hr after admission and he is now anuric.

Cr **3.4** (2.5 on admission, baseline 1.2) | FeNa 2.3%
Urinalysis: muddy brown casts
Kidney US: Normal

Likely ischemic acute tubular necrosis from prolonged hypoperfusion due to hypovolemia and sepsis

Monitor electrolytes, maintain euolemia, avoid overhydration (anuric), and avoid nephrotoxins.

Case 6

36yo man presents to the ED with diffuse muscle pain not responding to NSAIDs and dark urine two days after starting a new high intensity work-out program.

Cr **3.6** (baseline 1.2) | BUN 30 | FeNa 3% | CK 20,000
Urine dipstick: 3+ blood, no RBC on urinalysis
Kidney US: Normal

Likely tubular obstruction from myoglobin as a result of rhabdomyolysis PLUS injury 2/2 NSAIDs

Aggressive fluid resuscitation and discontinuation of NSAIDs

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Decreased Effective Arterial Blood Volume (EABV)

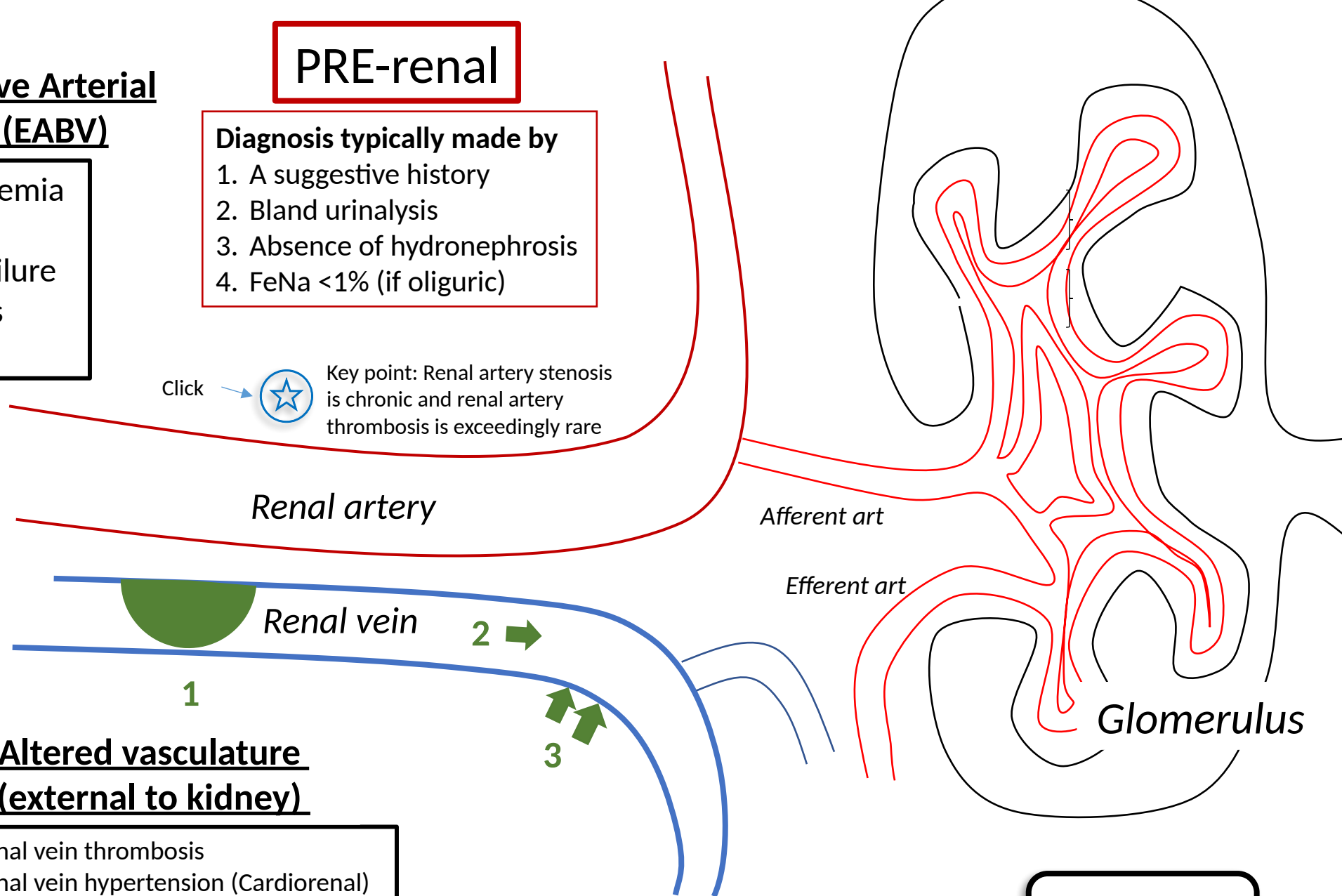
1. Hypovolemia
2. Sepsis
3. Heart failure
4. Cirrhosis
5. Burns

PRE-renal

Diagnosis typically made by

1. A suggestive history
2. Bland urinalysis
3. Absence of hydronephrosis
4. FeNa <1% (if oliguric)

Click  Key point: Renal artery stenosis is chronic and renal artery thrombosis is exceedingly rare



Altered vasculature (external to kidney)

1. Renal vein thrombosis
2. Renal vein hypertension (Cardiorenal)
3. Intra-Abdominal Hypertension (IAH)

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POST-renal

Diagnosis ruled out by:

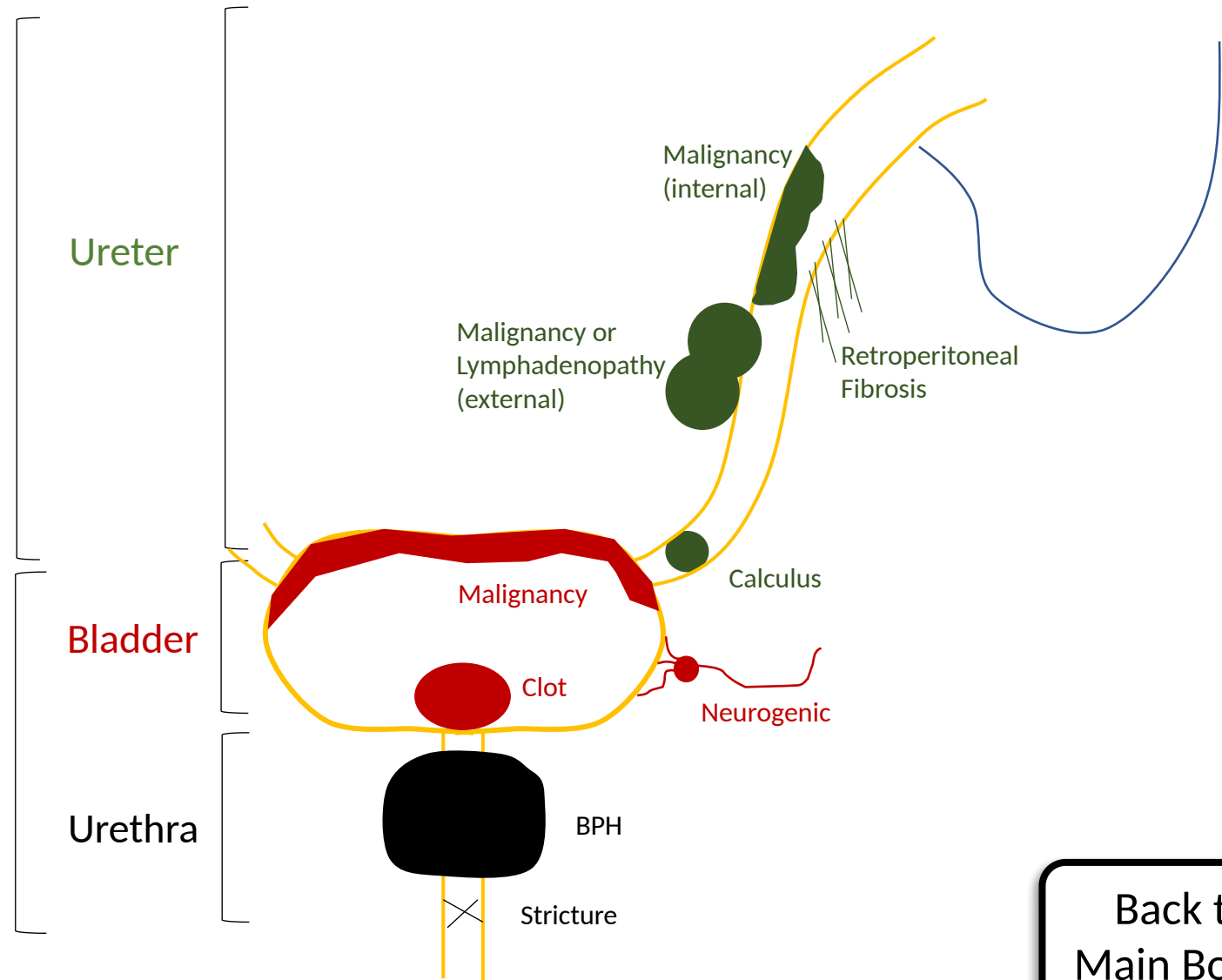
No hydronephrosis on an ultrasound of Kidney

If unilateral ≠ AKI

Key point: Ureteral obstruction will only lead to an AKI if obstruction is bilateral if the unilateral obstruction affecting single functional kidney



Bilateral obstruction = AKI



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